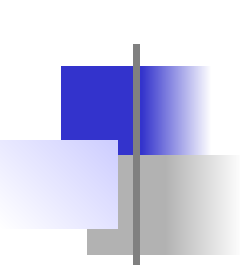


# The $z$ Transform

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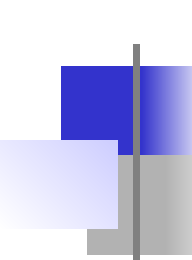
# Generalizing the DTFT

The forward DTFT is defined by  $X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$  in which

$\Omega$  is discrete-time radian frequency, a real variable. The quantity  $e^{j\Omega n}$  is then a complex sinusoid whose magnitude is always one and whose phase can range over all angles. It always lies on the unit circle in the complex plane. If we now replace  $e^{j\Omega}$  with a variable  $z$  that can

have any complex value we define the  $z$  transform  $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$ .

The DTFT expresses signals as linear combinations of complex sinusoids. The  $z$  transform expresses signals as linear combinations of complex exponentials.



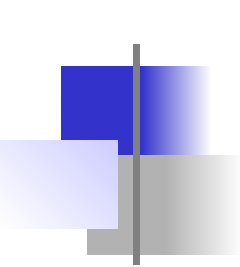
# Complex Exponential Excitation

Let the excitation of a discrete-time LTI system be a complex exponential of the form  $Az^n$  where  $z$  is, in general, complex and  $A$  is any constant. Using convolution, the response  $y[n]$  of an LTI system with impulse response  $h[n]$  to a complex exponential excitation  $x[n]$  is

$$y[n] = h[n] * Az^n = A \sum_{m=-\infty}^{\infty} h[m] z^{n-m} = \underbrace{Az^n}_{=x[n]} \sum_{m=-\infty}^{\infty} h[m] z^{-m}$$

The response is the product of the excitation and the  $z$  transform of

$$h[n] \text{ defined by } H(z) = \sum_{m=-\infty}^{\infty} h[m] z^{-m}.$$



# The Transfer Function

If an LTI system with impulse response  $h[n]$  is excited by a signal,  $x[n]$ , the  $z$  transform  $Y(z)$  of the response  $y[n]$  is

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n] z^{-n} = \sum_{n=-\infty}^{\infty} (h[n] * x[n]) z^{-n} = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} h[m] x[n-m] z^{-n}$$

$$Y(z) = \sum_{m=-\infty}^{\infty} h[m] \sum_{n=-\infty}^{\infty} x[n-m] z^{-n}$$

Let  $q = n - m$ . Then

$$Y(z) = \sum_{m=-\infty}^{\infty} h[m] \sum_{q=-\infty}^{\infty} x[q] z^{-(q+m)} = \underbrace{\sum_{m=-\infty}^{\infty} h[m] z^{-m}}_{=H(z)} \underbrace{\sum_{q=-\infty}^{\infty} x[q] z^{-q}}_{=X(z)}$$

$$Y(z) = H(z)X(z)$$

$H(z)$  is the **transfer function**.

# Systems Described by Difference Equations

The most common description of a discrete-time system is a difference equation of the general form

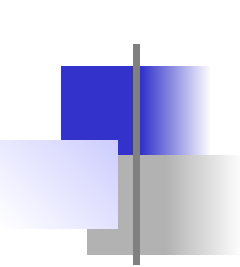
$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k].$$

It was shown in Chapter 5 that the transfer function for a system of this type is

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

or

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = z^{N-M} \frac{b_0 z^M + b_1 z^{M-1} + \cdots + b_{M-1} z + b_M}{a_0 z^N + a_1 z^{N-1} + \cdots + a_{N-1} z + a_N}$$



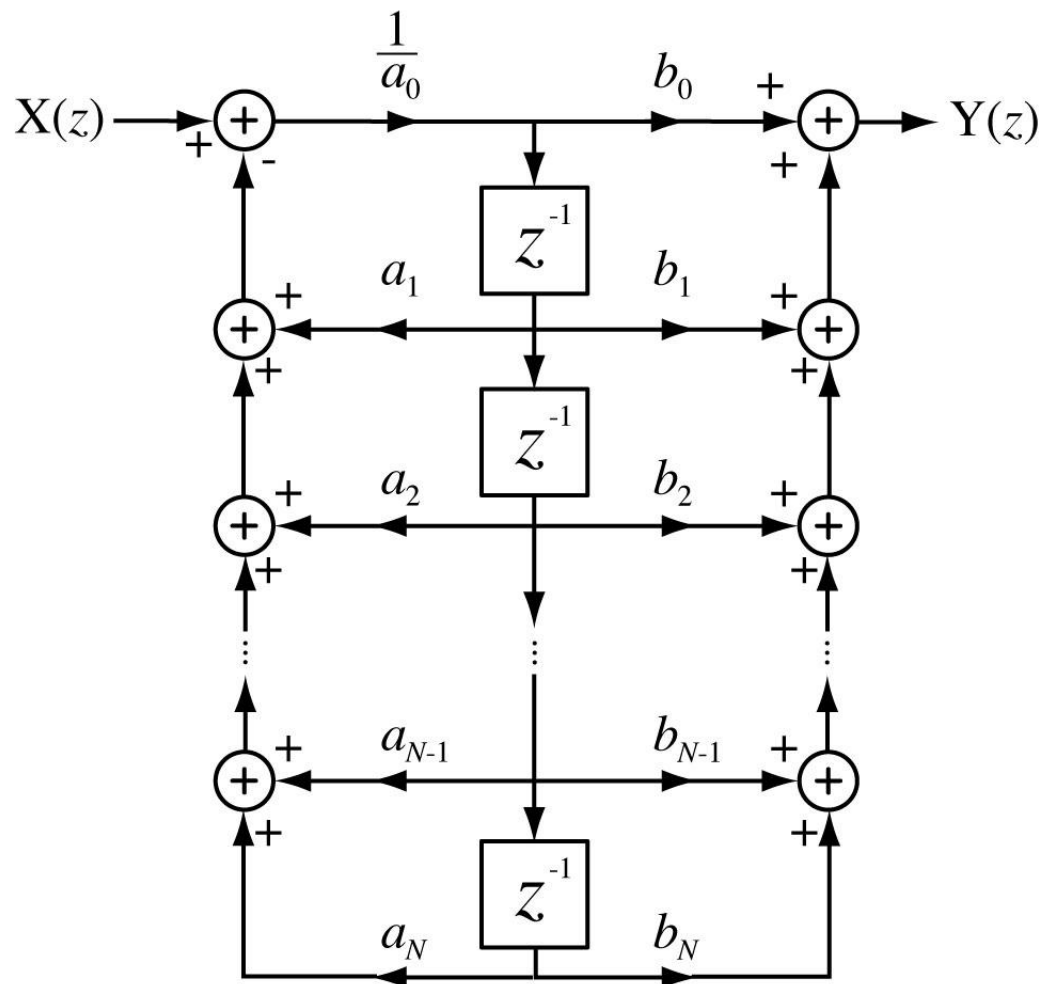
# Direct Form II Realization

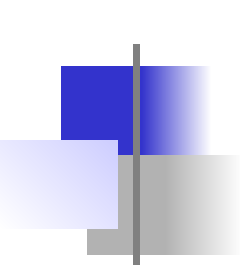
Direct Form II realization of a discrete-time system is similar in form to Direct Form II realization of continuous-time systems

A continuous-time system can be realized with integrators, summing junctions and multipliers

A discrete-time system can be realized with delays, summing junctions and multipliers

# Direct Form II Realization





# The Inverse $z$ Transform

The inversion integral is

$$x[n] = \frac{1}{j2\pi} \oint_C X(z) z^{n-1} dz.$$

This is a contour integral in the complex plane and is beyond the scope of this course. The notation  $x[n] \xleftrightarrow{z} X(z)$  indicates that  $x[n]$  and  $X(z)$  form a "z-transform pair".



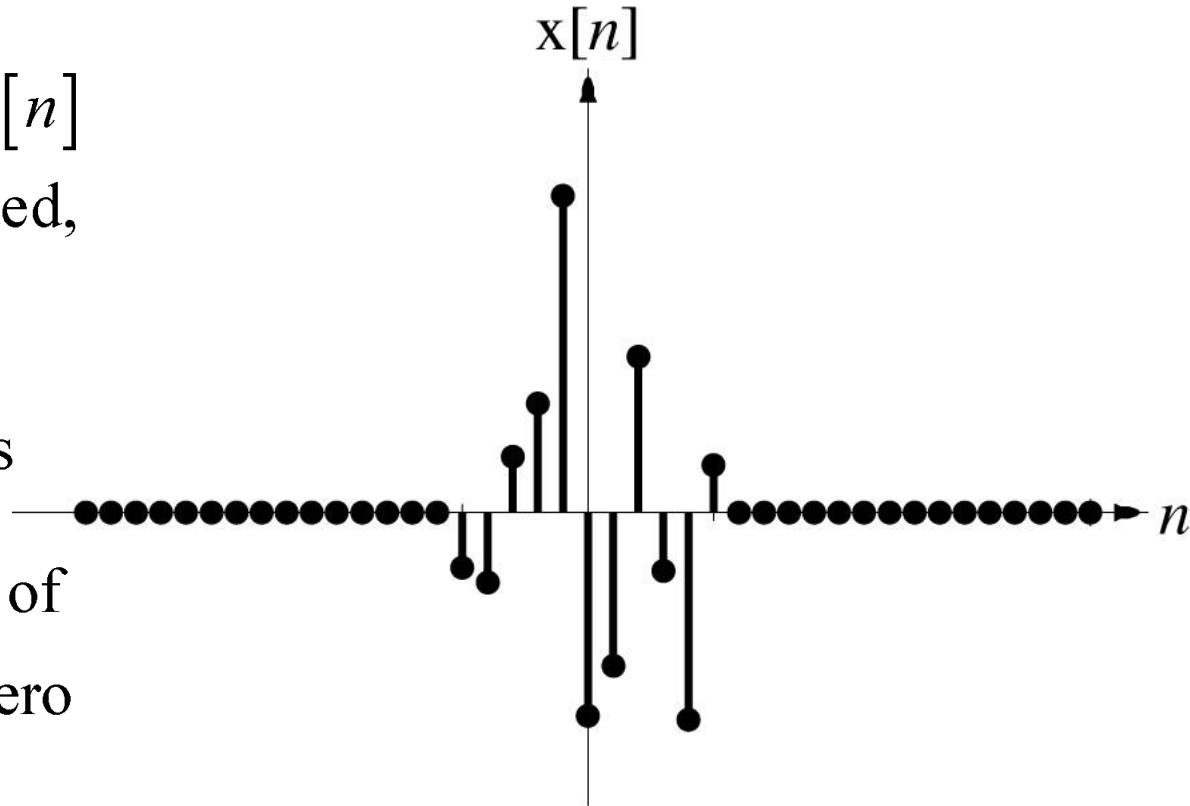
# Existence of the $z$ Transform

## Time Limited Signals

If a discrete-time signal  $x[n]$  is time limited and bounded, the  $z$  transformation

summation  $\sum_{n=-\infty}^{\infty} x[n]z^{-n}$  is

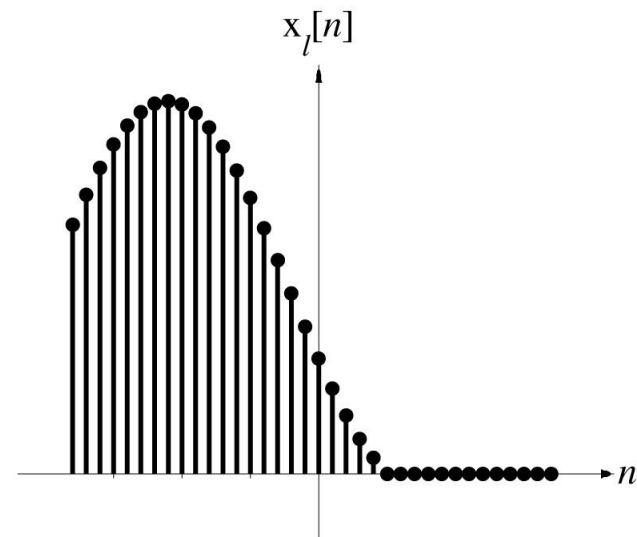
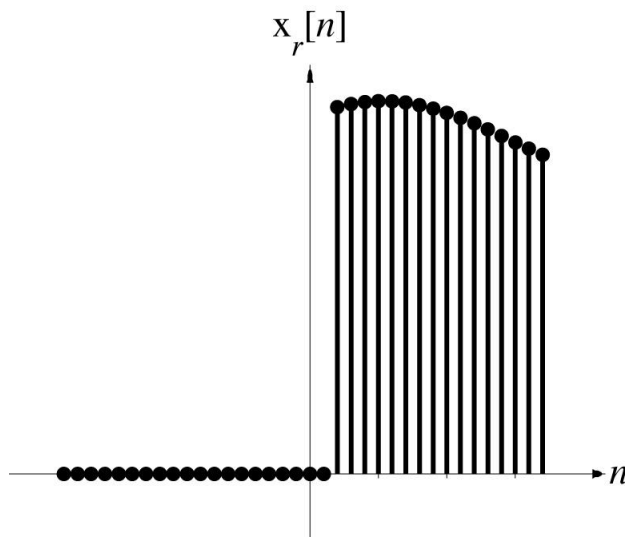
finite and the  $z$  transform of  $x[n]$  exists for any non-zero value of  $z$ .



# Existence of the $z$ Transform

## Right- and Left-Sided Signals

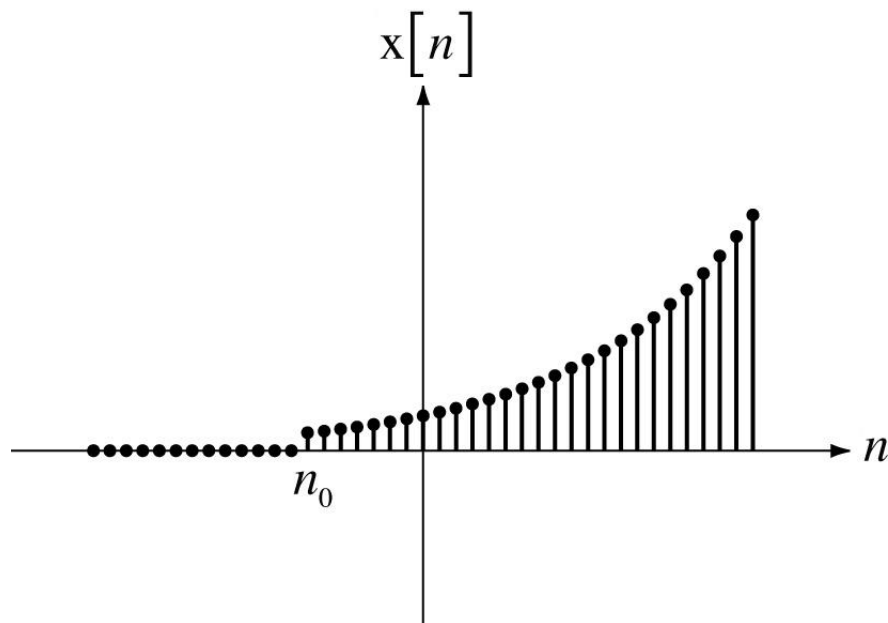
A right-sided signal  $x_r[n]$  is one for which  $x_r[n] = 0$  for any  $n < n_0$  and a left-sided signal  $x_l[n]$  is one for which  $x_l[n] = 0$  for any  $n > n_0$ .



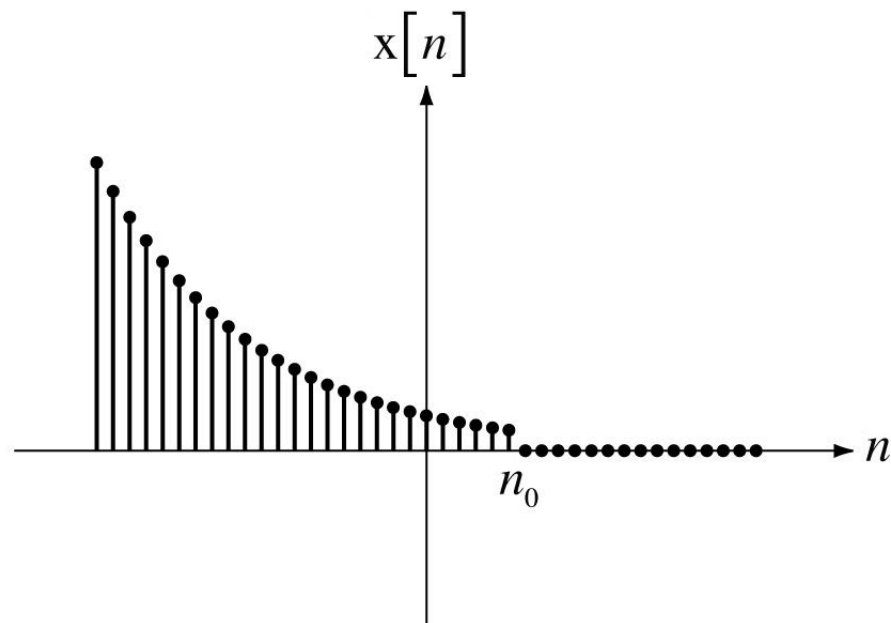
# Existence of the $z$ Transform

## Right- and Left-Sided Exponentials

$$x[n] = \alpha^n u[n - n_0] \quad , \quad \alpha \in \mathbb{R}$$



$$x[n] = \beta^n u[n_0 - n] \quad , \quad \beta \in \mathbb{R}$$



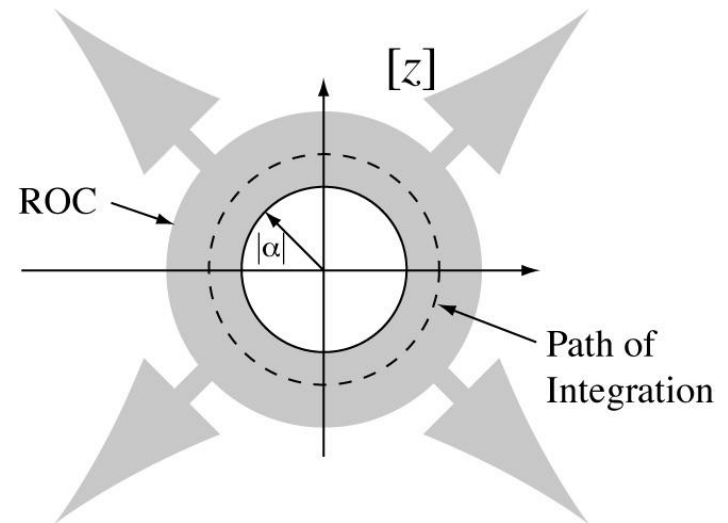
# Existence of the $z$ Transform

The  $z$  transform of  $x[n] = \alpha^n u[n - n_0]$  ,  $\alpha \in \mathbb{R}$  is

$$X(z) = \sum_{n=-\infty}^{\infty} \alpha^n u[n - n_0] z^{-n} = \sum_{n=n_0}^{\infty} (\alpha z^{-1})^n$$

if the series converges and it converges

if  $|z| > |\alpha|$ . The path of integration of the inverse  $z$  transform must lie in the region of the  $z$  plane outside a circle of radius  $|\alpha|$ .

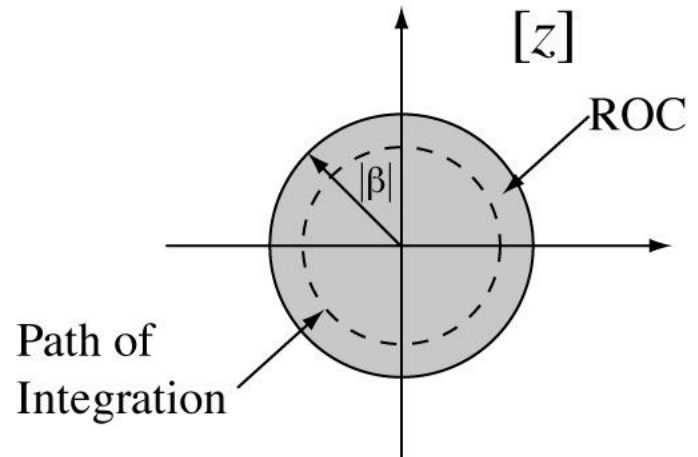


# Existence of the $z$ Transform

The  $z$  transform of  $x[n] = \beta^n u[n_0 - n]$ ,  $\beta \in \mathbb{R}$  is

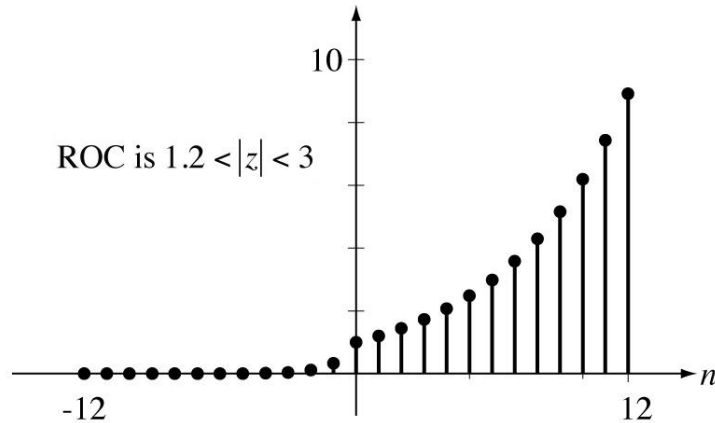
$$X(z) = \sum_{n=-\infty}^{n_0} \beta^n z^{-n} = \sum_{n=-\infty}^{n_0} (\beta z^{-1})^n = \sum_{n=-n_0}^{\infty} (\beta^{-1} z)^n$$

if the series converges and it converges if  $|z| < |\beta|$ . The path of integration of the inverse  $z$  transform must lie in the region of the  $z$  plane inside a circle of radius  $|\beta|$ .

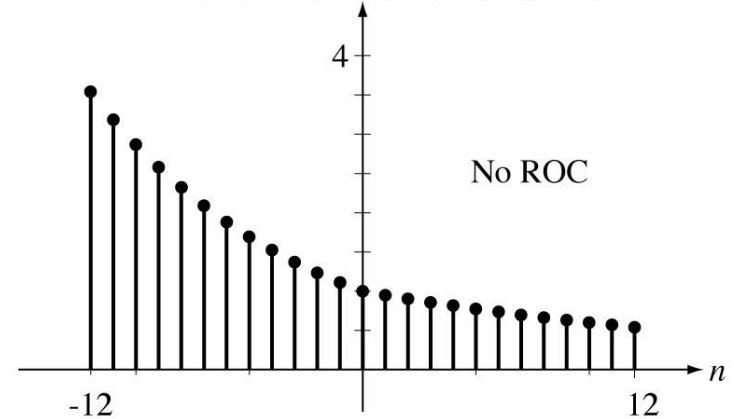


# Existence of the $z$ Transform

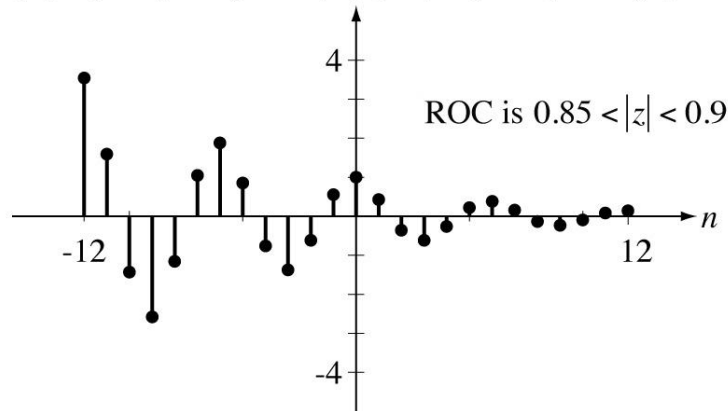
$$x[n] = (1.2)^n u[n] + (3)^n u[-n-1]$$



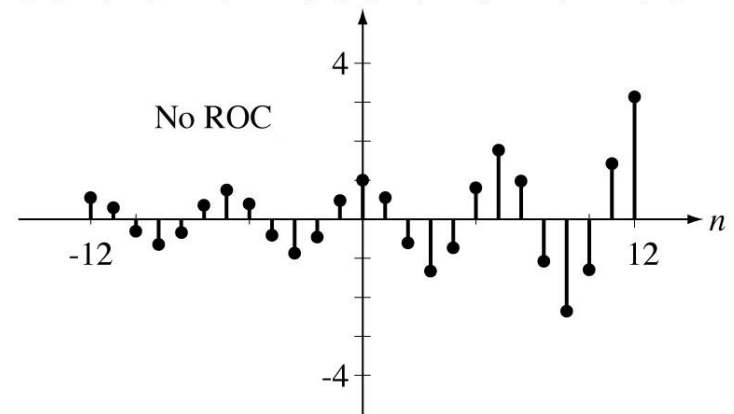
$$x[n] = (0.95)^n u[n] + (0.9)^n u[-n-1]$$



$$x[n] = (0.85)^n \cos(2\pi n/6) u[n] + (0.9)^n \cos(2\pi n/6) u[-n-1]$$



$$x[n] = (1.1)^n \cos(2\pi n/6) u[n] + (1.05)^n \cos(2\pi n/6) u[-n-1]$$



# Some Common $z$ Transform Pairs

$$\delta[n] \xleftrightarrow{z} 1, \text{ All } z$$

$$u[n] \xleftrightarrow{z} \frac{z}{z-1} = \frac{1}{1-z^{-1}}, \quad |z| > 1$$

$$-u[-n-1] \xleftrightarrow{z} \frac{z}{z-1}, \quad |z| < 1$$

$$\alpha^n u[n] \xleftrightarrow{z} \frac{z}{z-\alpha} = \frac{1}{1-\alpha z^{-1}}, \quad |z| > |\alpha|$$

$$-\alpha^n u[-n-1] \xleftrightarrow{z} \frac{z}{z-\alpha} = \frac{1}{1-\alpha z^{-1}}, \quad |z| < |\alpha|$$

$$n u[n] \xleftrightarrow{z} \frac{z}{(z-1)^2} = \frac{z^{-1}}{(1-z^{-1})^2}, \quad |z| > 1$$

$$-n u[-n-1] \xleftrightarrow{z} \frac{z}{(z-1)^2} = \frac{z^{-1}}{(1-z^{-1})^2}, \quad |z| < 1$$

$$n \alpha^n u[n] \xleftrightarrow{z} \frac{\alpha z}{(z-\alpha)^2} = \frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}, \quad |z| > |\alpha|$$

$$-n \alpha^n u[-n-1] \xleftrightarrow{z} \frac{\alpha z}{(z-\alpha)^2} = \frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}, \quad |z| < |\alpha|$$

$$\sin(\Omega_0 n) u[n] \xleftrightarrow{z} \frac{z \sin(\Omega_0)}{z^2 - 2z \cos(\Omega_0) + 1}, \quad |z| > 1$$

$$-\sin(\Omega_0 n) u[-n-1] \xleftrightarrow{z} \frac{z \sin(\Omega_0)}{z^2 - 2z \cos(\Omega_0) + 1}, \quad |z| < 1$$

$$\cos(\Omega_0 n) u[n] \xleftrightarrow{z} \frac{z[z - \cos(\Omega_0)]}{z^2 - 2z \cos(\Omega_0) + 1}, \quad |z| > 1$$

$$-\cos(\Omega_0 n) u[-n-1] \xleftrightarrow{z} \frac{z[z - \cos(\Omega_0)]}{z^2 - 2z \cos(\Omega_0) + 1}, \quad |z| < 1$$

$$\alpha^n \sin(\Omega_0 n) u[n] \xleftrightarrow{z} \frac{z \alpha \sin(\Omega_0)}{z^2 - 2\alpha z \cos(\Omega_0) + \alpha^2}, \quad |z| > |\alpha|, \quad -\alpha^n \sin(\Omega_0 n) u[-n-1] \xleftrightarrow{z} \frac{z \alpha \sin(\Omega_0)}{z^2 - 2\alpha z \cos(\Omega_0) + \alpha^2}, \quad |z| < |\alpha|$$

$$\alpha^n \cos(\Omega_0 n) u[n] \xleftrightarrow{z} \frac{z[z - \alpha \cos(\Omega_0)]}{z^2 - 2\alpha z \cos(\Omega_0) + \alpha^2}, \quad |z| > |\alpha|, \quad -\alpha^n \cos(\Omega_0 n) u[-n-1] \xleftrightarrow{z} \frac{z[z - \alpha \cos(\Omega_0)]}{z^2 - 2\alpha z \cos(\Omega_0) + \alpha^2}, \quad |z| < |\alpha|$$

$$\alpha^{|n|} \xleftrightarrow{z} \frac{z}{z-\alpha} - \frac{z}{z-\alpha^{-1}}, \quad |\alpha| < |z| < |\alpha|^{-1}$$

$$u[n-n_0] - u[n-n_1] \xleftrightarrow{z} \frac{z}{z-1} (z^{-n_0} - z^{-n_1}) = \frac{z^{n_1-n_0-1} + z^{n_1-n_0-2} + \dots + z + 1}{z^{n_1-1}}, \quad |z| > 0$$

# z-Transform Properties

Given the z-transform pairs  $g[n] \xleftrightarrow{z} G(z)$  and  $h[n] \xleftrightarrow{z} H(z)$  with ROC's of  $\text{ROC}_G$  and  $\text{ROC}_H$  respectively the following properties apply to the  $z$  transform.

## Linearity

$$\alpha g[n] + \beta h[n] \xleftrightarrow{z} \alpha G(z) + \beta H(z)$$

$$\text{ROC} = \text{ROC}_G \cap \text{ROC}_H$$

## Time Shifting

$$g[n - n_0] \xleftrightarrow{z} z^{-n_0} G(z)$$

$$\text{ROC} = \text{ROC}_G \text{ except perhaps } z = 0 \text{ or } z \rightarrow \infty$$

## Change of Scale in $z$

$$\alpha^n g[n] \xleftrightarrow{z} G(z / \alpha)$$

$$\text{ROC} = |\alpha| \text{ROC}_G$$



# z-Transform Properties

## Time Reversal

$$g[-n] \xleftrightarrow{z} G(z^{-1})$$

$$\text{ROC} = 1 / \text{ROC}_G$$

## Time Expansion

$$\left\{ \begin{array}{ll} g[n/k] & , n/k \text{ and integer} \\ 0 & , \text{otherwise} \end{array} \right\} \xleftrightarrow{z} G(z^k)$$

$$\text{ROC} = (\text{ROC}_G)^{1/k}$$

## Conjugation

$$g^*[n] \xleftrightarrow{z} G^*(z^*)$$

$$\text{ROC} = \text{ROC}_G$$

## z-Domain Differentiation

$$-n g[n] \xleftrightarrow{z} z \frac{d}{dz} G(z)$$

$$\text{ROC} = \text{ROC}_G$$

# z-Transform Properties

## Convolution

$$g[n] * h[n] \xleftrightarrow{Z} H(z)G(z)$$

## First Backward Difference

$$g[n] - g[n-1] \xleftrightarrow{Z} (1 - z^{-1})G(z)$$

$$\text{ROC} \supseteq \text{ROC}_G \cap |z| > 0$$

## Accumulation

$$\sum_{m=-\infty}^n g[m] \xleftrightarrow{Z} \frac{z}{z-1} G(z)$$

$$\text{ROC} \supseteq \text{ROC}_G \cap |z| > 1$$

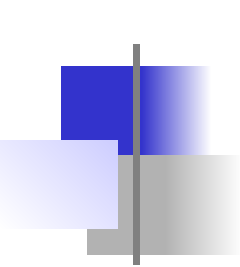
## Initial Value Theorem

$$\text{If } g[n] = 0, n < 0 \text{ then } g[0] = \lim_{z \rightarrow \infty} G(z)$$

## Final Value Theorem

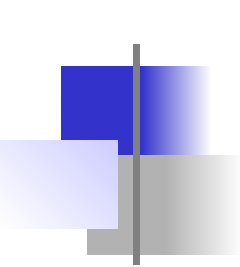
$$\text{If } g[n] = 0, n < 0, \lim_{n \rightarrow \infty} g[n] = \lim_{z \rightarrow 1} (z-1)G(z)$$

if  $\lim_{n \rightarrow \infty} g[n]$  exists.



# $z$ -Transform Properties

For the final-value theorem to apply to a function  $G(z)$  all the finite poles of the function  $(z - 1)G(z)$  must lie in the open interior of the unit circle of the  $z$  plane. Notice this does not say that all the poles of  $G(z)$  must lie in the open interior of the unit circle.  $G(z)$  could have a single pole at  $z = 1$  and the final-value theorem could still apply.



# The Inverse $z$ Transform

## Synthetic Division

For rational  $z$  transforms of the form

$$H(z) = \frac{b_M z^M + b_{M-1} z^{M-1} + \dots + b_1 z + b_0}{a_N z^N + a_{N-1} z^{N-1} + \dots + a_1 z + a_0}$$

we can always find the inverse  $z$  transform by synthetic division. For example,

$$H(z) = \frac{(z - 1.2)(z + 0.7)(z + 0.4)}{(z - 0.2)(z - 0.8)(z + 0.5)}, \quad |z| > 0.8$$

$$H(z) = \frac{z^3 - 0.1z^2 - 1.04z - 0.336}{z^3 - 0.5z^2 - 0.34z + 0.08}, \quad |z| > 0.8$$

# The Inverse $z$ Transform

## Synthetic Division

$$\begin{array}{r}
 \phantom{z^3 - 0.5z^2 - 0.34z + 0.08} \overline{1 + 0.4z^{-1} + 0.5z^{-2} \dots} \\
 z^3 - 0.5z^2 - 0.34z + 0.08 \bigg) z^3 - 0.1z^2 - 1.04z - 0.336 \\
 \underline{z^3 - 0.5z^2 - 0.34z + 0.08} \\
 0.4z^2 - 0.7z - 0.256 \\
 \underline{0.4z^2 - 0.2z - 0.136 - 0.032z^{-1}} \\
 0.5z - 0.12 + 0.032z^{-1} \\
 \vdots \quad \quad \quad \vdots
 \end{array}$$

The inverse  $z$  transform is

$$\delta[n] + 0.4\delta[n-1] + 0.5\delta[n-2] \dots \xleftrightarrow{z} 1 + 0.4z^{-1} + 0.5z^{-2} \dots$$

# The Inverse $z$ Transform

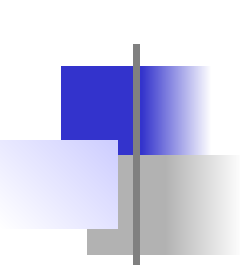
## Synthetic Division

We could have done the synthetic division this way.

$$\begin{array}{r}
 \phantom{0.08 - 0.34z - 0.5z^2 + z^3} \overline{-4.2 - 30.85z - 158.613z^2 \dots} \\
 0.08 - 0.34z - 0.5z^2 + z^3 \bigg) \phantom{-} -0.336 - 1.04z - 0.1z^2 + z^3 \\
 \phantom{0.08 - 0.34z - 0.5z^2 + z^3} \underline{-0.336 + 1.428z + 2.1z^2 - 4.2z^3} \\
 \phantom{0.08 - 0.34z - 0.5z^2 + z^3} \phantom{-} -2.468z - 2.2z^2 + 5.2z^3 \\
 \phantom{0.08 - 0.34z - 0.5z^2 + z^3} \phantom{-} \underline{-2.468z + 10.489z^2 + 15.425z^3 - 30.85z^4} \\
 \phantom{0.08 - 0.34z - 0.5z^2 + z^3} \phantom{-} \phantom{-} -12.689z^2 - 10.225z^3 + 30.85z^4 \\
 \phantom{0.08 - 0.34z - 0.5z^2 + z^3} \phantom{-} \phantom{-} \phantom{-} \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots
 \end{array}$$

$$-4.2\delta[n] - 30.85\delta[n+1] - 158.613\delta[n+2] \dots \xleftrightarrow{z} -4.2 - 30.85z - 158.613z^2 \dots$$

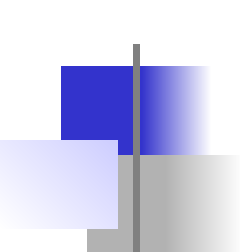
but with the restriction  $|z| > 0.8$  this second form does not converge and is therefore not the inverse  $z$  transform.



# The Inverse $z$ Transform

## Synthetic Division

We can always find the inverse  $z$  transform of a rational function with synthetic division but the result is not in closed form. In most practical cases a closed-form solution is preferred.



# Partial Fraction Expansion

Partial-fraction expansion works for inverse  $z$  transforms the same way it does for inverse Laplace transforms. But there is a situation that is quite common in inverse  $z$  transforms which deserves mention. It is very common to have  $z$ -domain functions in which the number of finite zeros equals the number of finite poles (making the expression improper in  $z$ ) with at least one zero at  $z = 0$ .

$$H(z) = \frac{z^{N-M} (z - z_1)(z - z_2) \cdots (z - z_M)}{(z - p_1)(z - p_2) \cdots (z - p_N)}$$



# Partial Fraction Expansion

Dividing both sides by  $z$  we get

$$\frac{H(z)}{z} = \frac{z^{N-M-1}(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)}$$

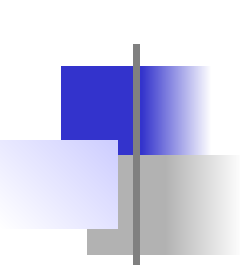
and the fraction on the right is now proper in  $z$  and can be expanded in partial fractions.

$$\frac{H(z)}{z} = \frac{K_1}{z-p_1} + \frac{K_2}{z-p_2} + \cdots + \frac{K_N}{z-p_N}$$

Then both sides can be multiplied by  $z$  and the inverse transform can be found.

$$H(z) = \frac{zK_1}{z-p_1} + \frac{zK_2}{z-p_2} + \cdots + \frac{zK_N}{z-p_N}$$

$$h[n] = K_1 p_1^n u[n] + K_2 p_2^n u[n] + \cdots + K_N p_N^n u[n]$$



# $z$ -Transform Properties

An LTI system has a transfer function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z - 1/2}{z^2 - z + 2/9}, \quad |z| > 2/3$$

Using the time-shifting property of the  $z$  transform draw a block diagram realization of the system.

$$Y(z)(z^2 - z + 2/9) = X(z)(z - 1/2)$$

$$z^2 Y(z) = z X(z) - (1/2) X(z) + z Y(z) - (2/9) Y(z)$$

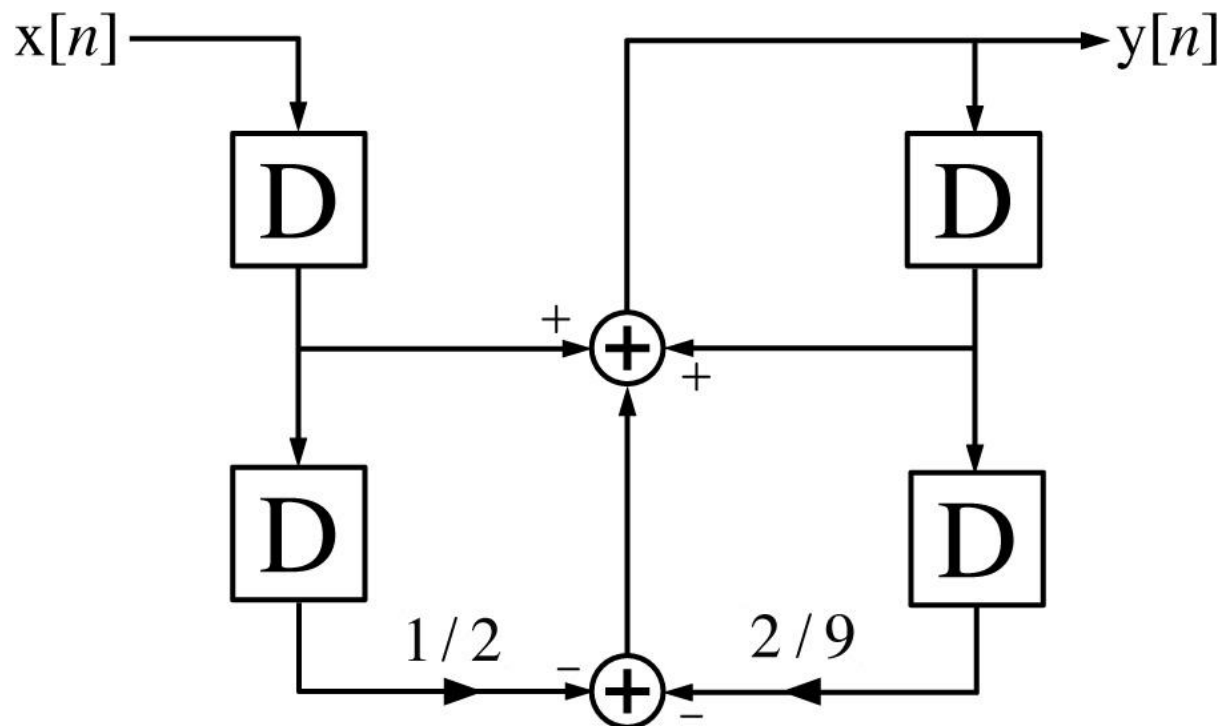
$$Y(z) = z^{-1} X(z) - (1/2) z^{-2} X(z) + z^{-1} Y(z) - (2/9) z^{-2} Y(z)$$

# $z$ -Transform Properties

$$Y(z) = z^{-1} X(z) - (1/2)z^{-2} X(z) + z^{-1} Y(z) - (2/9)z^{-2} Y(z)$$

Using the time-shifting property

$$y[n] = x[n-1] - (1/2)x[n-2] + y[n-1] - (2/9)y[n-2]$$



# z-Transform Properties

Let  $g[n] \xleftrightarrow{z} G(z) = \frac{z-1}{(z-0.8e^{-j\pi/4})(z-0.8e^{+j\pi/4})}$ . Draw a

pole-zero diagram for  $G(z)$  and for the  $z$  transform of  $e^{j\pi n/8}g[n]$ .

The poles of  $G(z)$  are at  $z = 0.8e^{\pm j\pi/4}$  and its single finite zero is at  $z = 1$ . Using the change of scale property

$$e^{j\pi n/8}g[n] \xleftrightarrow{z} G(ze^{-j\pi/8}) = \frac{ze^{-j\pi/8} - 1}{(ze^{-j\pi/8} - 0.8e^{-j\pi/4})(ze^{-j\pi/8} - 0.8e^{+j\pi/4})}$$

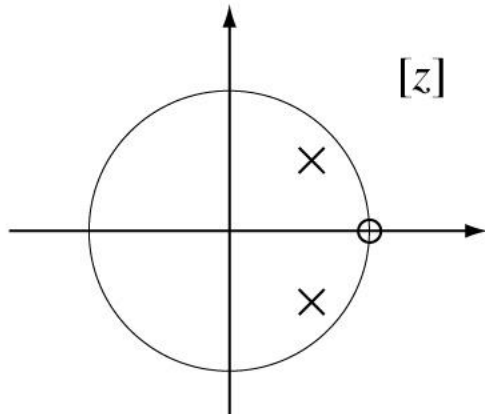
$$G(ze^{-j\pi/8}) = \frac{e^{-j\pi/8}(z - e^{j\pi/8})}{e^{-j\pi/8}(z - 0.8e^{-j\pi/8})e^{-j\pi/8}(z - 0.8e^{+j3\pi/8})}$$

$$G(ze^{-j\pi/8}) = e^{j\pi/8} \frac{z - e^{j\pi/8}}{(z - 0.8e^{-j\pi/8})(z - 0.8e^{+j3\pi/8})}$$

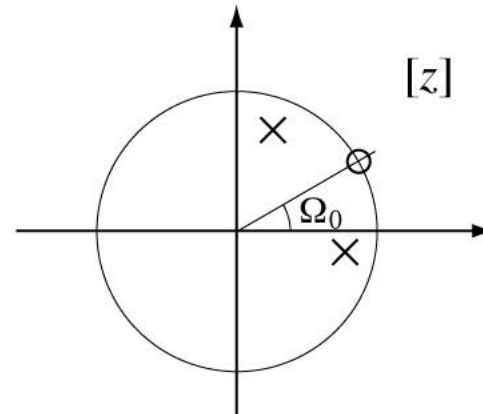
# z-Transform Properties

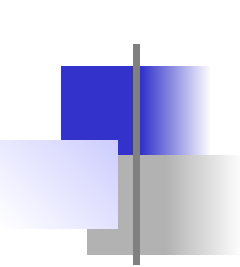
$G(z e^{-j\pi/8})$  has poles at  $z = 0.8e^{-j\pi/8}$  and  $0.8e^{+j3\pi/8}$  and a zero at  $z = e^{j\pi/8}$ . All the finite zero and pole locations have been rotated in the  $z$  plane by  $\pi/8$  radians.

Pole-zero Plot of  $G(z)$



Pole-zero Plot of  $G(z e^{-j\Omega_0})$





# $z$ -Transform Properties

Using the accumulation property and  $u[n] \xleftrightarrow{z} \frac{z}{z-1}$  ,  $|z| > 1$

show that the  $z$  transform of  $nu[n]$  is  $\frac{z}{(z-1)^2}$  ,  $|z| > 1$ .

$$nu[n] = \sum_{m=0}^n u[m-1]$$

$$u[n-1] \xleftrightarrow{z} z^{-1} \frac{z}{z-1} = \frac{1}{z-1} , \quad |z| > 1$$

$$nu[n] = \sum_{m=0}^n u[m-1] \xleftrightarrow{z} \left( \frac{z}{z-1} \right) \frac{1}{z-1} = \frac{z}{(z-1)^2} , \quad |z| > 1$$

# Inverse $z$ Transform Example

Find the inverse  $z$  transform of

$$X(z) = \frac{z}{z-0.5} - \frac{z}{z+2}, \quad 0.5 < |z| < 2$$

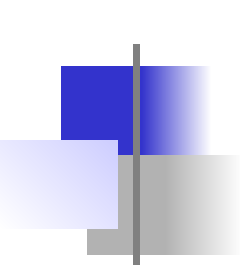
Right-sided signals have ROC's that are outside a circle and left-sided signals have ROC's that are inside a circle. Using

$$\alpha^n u[n] \xleftrightarrow{z} \frac{z}{z-\alpha} = \frac{1}{1-\alpha z^{-1}}, \quad |z| > |\alpha|$$

$$-\alpha^n u[-n-1] \xleftrightarrow{z} \frac{z}{z-\alpha} = \frac{1}{1-\alpha z^{-1}}, \quad |z| < |\alpha|$$

We get

$$(0.5)^n u[n] + (-2)^n u[-n-1] \xleftrightarrow{z} X(z) = \frac{z}{z-0.5} - \frac{z}{z+2}, \quad 0.5 < |z| < 2$$



# Inverse $z$ Transform Example

Find the inverse  $z$  transform of

$$X(z) = \frac{z}{z - 0.5} - \frac{z}{z + 2}, \quad |z| > 2$$

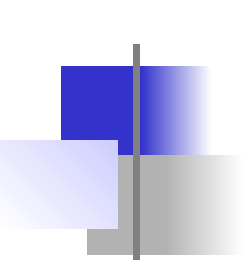
In this case, both signals are right sided. Then using

$$\alpha^n u[n] \xleftrightarrow{z} \frac{z}{z - \alpha} = \frac{1}{1 - \alpha z^{-1}}, \quad |z| > |\alpha|$$

We get

$$\left[ (0.5)^n - (-2)^n \right] u[n] \xleftrightarrow{z} X(z) = \frac{z}{z - 0.5} - \frac{z}{z + 2}, \quad |z| > 2$$





# Inverse $z$ Transform Example

Find the inverse  $z$  transform of

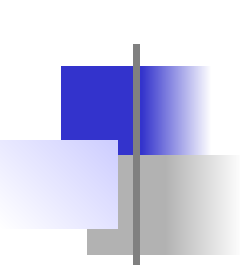
$$X(z) = \frac{z}{z-0.5} - \frac{z}{z+2}, \quad |z| < 0.5$$

In this case, both signals are left sided. Then using

$$-\alpha^n u[-n-1] \xleftrightarrow{z} \frac{z}{z-\alpha} = \frac{1}{1-\alpha z^{-1}}, \quad |z| < |\alpha|$$

We get

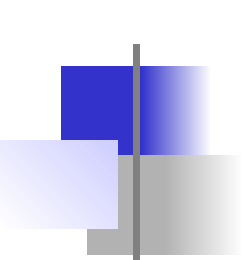
$$-\left[(0.5)^n - (-2)^n\right] u[-n-1] \xleftrightarrow{z} X(z) = \frac{z}{z-0.5} - \frac{z}{z+2}, \quad |z| < 0.5$$



# The Unilateral $z$ Transform

Just as it was convenient to define a unilateral Laplace transform it is convenient for analogous reasons to define a unilateral  $z$  transform

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$



# Properties of the Unilateral $z$ Transform

If two causal discrete-time signals form these transform pairs,

$g[n] \xleftrightarrow{z} G(z)$  and  $h[n] \xleftrightarrow{z} H(z)$  then the following properties hold for the unilateral  $z$  transform.

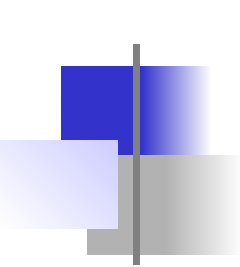
## Time Shifting

Delay:  $g[n - n_0] \xleftrightarrow{z} z^{-n_0} G(z), n_0 \geq 0$

Advance:  $g[n + n_0] \xleftrightarrow{z} z^{n_0} \left( G(z) - \sum_{m=0}^{n_0-1} g[m] z^{-m} \right), n_0 > 0$

## Accumulation:

$$\sum_{m=0}^n g[m] \xleftrightarrow{z} \frac{z}{z-1} G(z)$$



# Solving Difference Equations

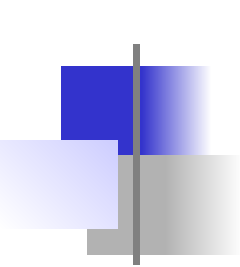
The unilateral z transform is well suited to solving difference equations with initial conditions. For example,

$$y[n+2] - \frac{3}{2}y[n+1] + \frac{1}{2}y[n] = (1/4)^n, \text{ for } n \geq 0$$
$$y[0] = 10 \text{ and } y[1] = 4$$

z transforming both sides,

$$z^2 [Y(z) - y[0] - z^{-1}y[1]] - \frac{3}{2}z[Y(z) - y[0]] + \frac{1}{2}Y(z) = \frac{z}{z - 1/4}$$

the initial conditions are called for systematically.



# Solving Difference Equations

Applying initial conditions and solving,

$$Y(z) = z \left( \frac{16/3}{z - 1/4} + \frac{4}{z - 1/2} + \frac{2/3}{z - 1} \right)$$

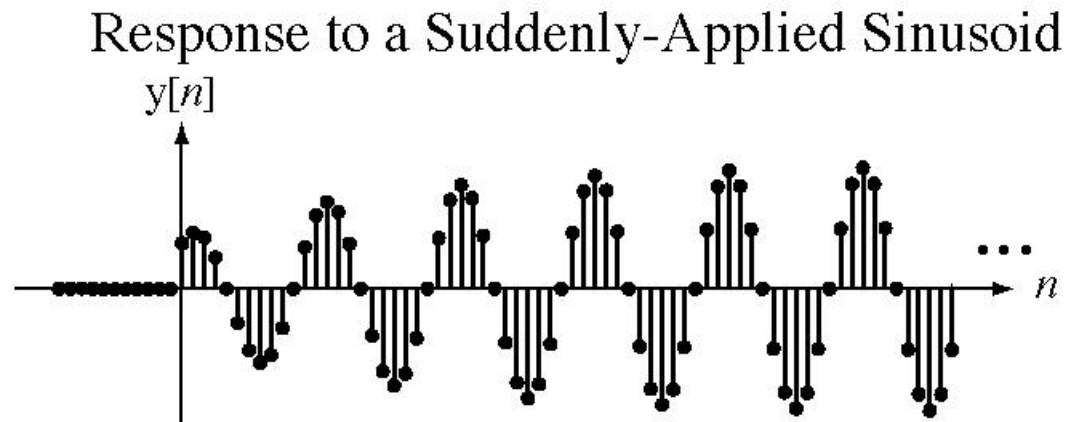
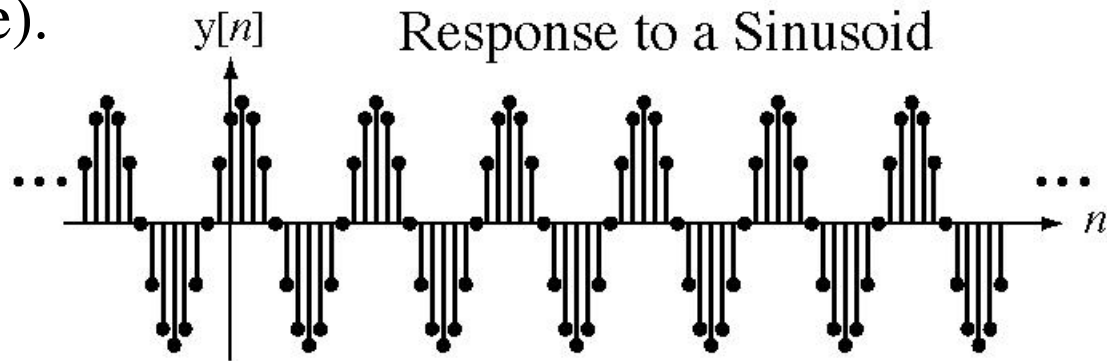
and

$$y[n] = \left[ \frac{16}{3} \left( \frac{1}{4} \right)^n + 4 \left( \frac{1}{2} \right)^n + \frac{2}{3} \right] u[n]$$

This solution satisfies the difference equation and the initial conditions.

# Pole-Zero Diagrams and Frequency Response

For a stable system, the response to a sinusoid applied at time  $t = 0$  approaches the response to a true sinusoid (applied for all time).



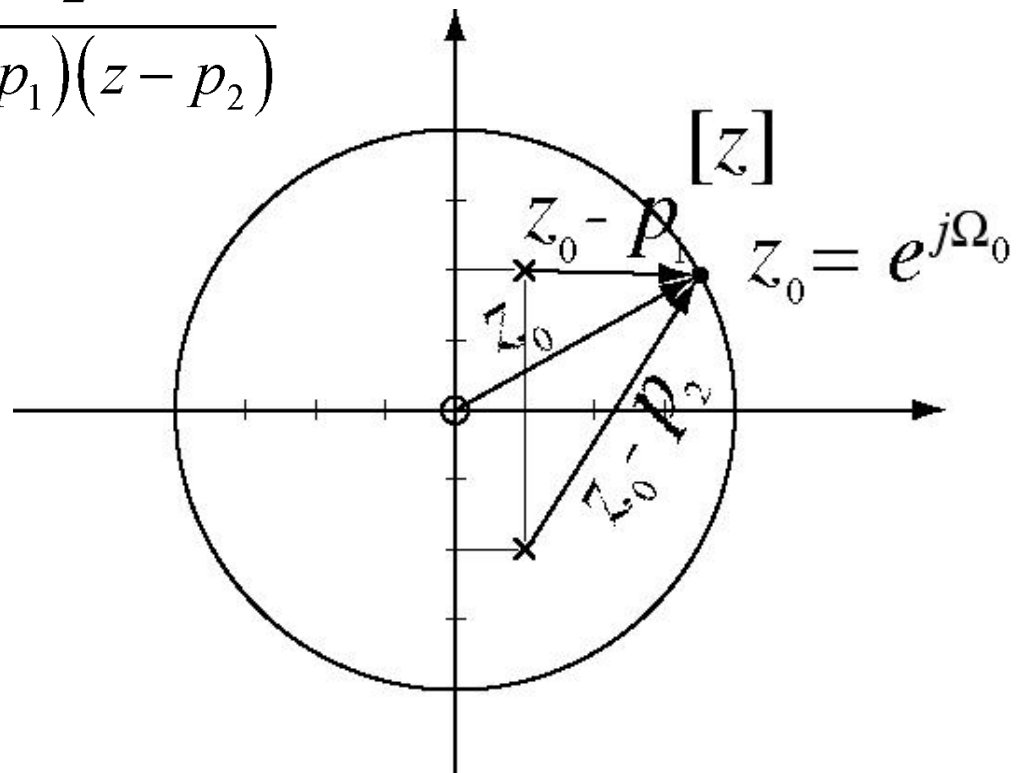
# Pole-Zero Diagrams and Frequency Response

Let the transfer function of a system be

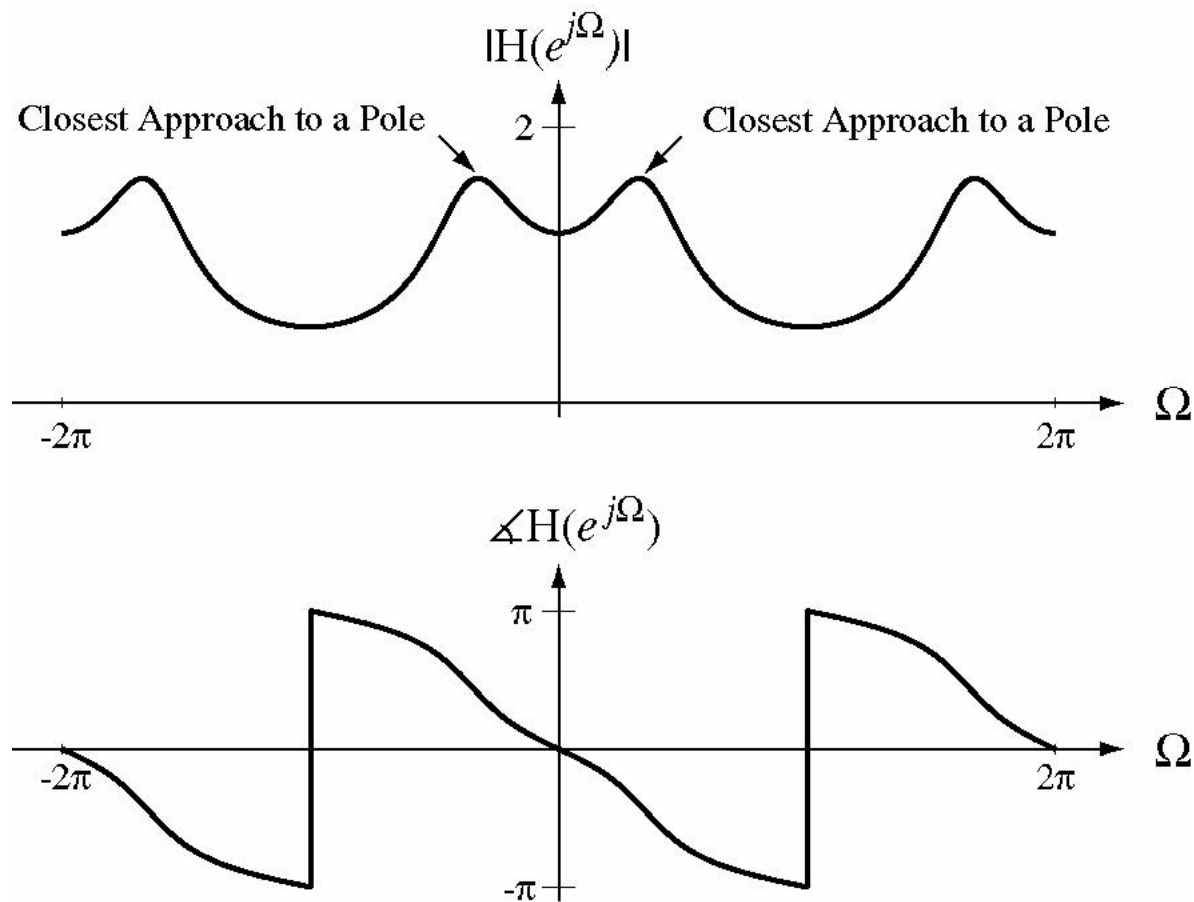
$$H(z) = \frac{z}{z^2 - z/2 + 5/16} = \frac{z}{(z - p_1)(z - p_2)}$$

$$p_1 = \frac{1+j2}{4}, \quad p_2 = \frac{1-j2}{4}$$

$$|H(e^{j\Omega})| = \frac{|e^{j\Omega}|}{|e^{j\Omega} - p_1| |e^{j\Omega} - p_2|}$$



# Pole-Zero Diagrams and Frequency Response







# Transform Method Comparison

A system with transfer function  $H(z) = \frac{z}{(z - 0.3)(z + 0.8)}$ ,  $|z| > 0.8$

is excited by a unit sequence. Find the total response.

Using  $z$ -transform methods,

$$Y(z) = H(z)X(z) = \frac{z}{(z - 0.3)(z + 0.8)} \times \frac{z}{z - 1}, \quad |z| > 1$$

$$Y(z) = \frac{z^2}{(z - 0.3)(z + 0.8)(z - 1)} = -\frac{0.1169}{z - 0.3} + \frac{0.3232}{z + 0.8} + \frac{0.7937}{z - 1}, \quad |z| > 1$$

$$y[n] = \left[ -0.1169(0.3)^{n-1} + 0.3232(-0.8)^{n-1} + 0.7937 \right] u[n-1]$$



# Transform Method Comparison

Using the DTFT,

$$H(e^{j\Omega}) = \frac{e^{j\Omega}}{(e^{j\Omega} - 0.3)(e^{j\Omega} + 0.8)}$$

$$Y(e^{j\Omega}) = H(e^{j\Omega})X(e^{j\Omega}) = \frac{e^{j\Omega}}{(e^{j\Omega} - 0.3)(e^{j\Omega} + 0.8)} \times \underbrace{\left( \frac{1}{1 - e^{-j\Omega}} + \pi \delta_{2\pi}(\Omega) \right)}_{\text{DTFT of a Unit Sequence}}$$

$$Y(e^{j\Omega}) = \frac{e^{j2\Omega}}{(e^{j\Omega} - 0.3)(e^{j\Omega} + 0.8)(e^{j\Omega} - 1)} + \pi \frac{e^{j\Omega}}{(e^{j\Omega} - 0.3)(e^{j\Omega} + 0.8)} \delta_{2\pi}(\Omega)$$

$$Y(e^{j\Omega}) = \frac{-0.1169}{e^{j\Omega} - 0.3} + \frac{0.3232}{e^{j\Omega} + 0.8} + \frac{0.7937}{e^{j\Omega} - 1} + \frac{\pi}{(1 - 0.3)(1 + 0.8)} \delta_{2\pi}(\Omega)$$

# Transform Method Comparison

Using the equivalence property of the impulse and the periodicity of both  $\delta_{2\pi}(\Omega)$  and  $e^{j\Omega}$

$$Y(e^{j\Omega}) = \frac{-0.1169e^{-j\Omega}}{1-0.3e^{-j\Omega}} + \frac{0.3232e^{-j\Omega}}{1+0.8e^{-j\Omega}} + \frac{0.7937e^{-j\Omega}}{1-e^{-j\Omega}} + 2.4933\delta_{2\pi}(\Omega)$$

Then, manipulating this expression into a form for which the inverse DTFT is direct

$$Y(e^{j\Omega}) = \frac{-0.1169e^{-j\Omega}}{1-0.3e^{-j\Omega}} + \frac{0.3232e^{-j\Omega}}{1+0.8e^{-j\Omega}} + 0.7937 \left( \frac{e^{-j\Omega}}{1-e^{-j\Omega}} + \pi\delta_{2\pi}(\Omega) \right) \\ \underbrace{-0.7937\pi\delta_{2\pi}(\Omega) + 2.4933\delta_{2\pi}(\Omega)}_{=0}$$



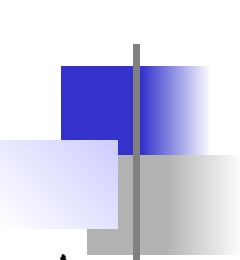
# Transform Method Comparison

$$Y(e^{j\Omega}) = \frac{-0.1169e^{-j\Omega}}{1 - 0.3e^{-j\Omega}} + \frac{0.3232e^{-j\Omega}}{1 + 0.8e^{-j\Omega}} + 0.7937 \left( \frac{e^{-j\Omega}}{1 - e^{-j\Omega}} + \pi\delta_{2\pi}(\Omega) \right)$$

Finding the inverse DTFT,

$$y[n] = \left[ -0.1169(0.3)^{n-1} + 0.3232(-0.8)^{n-1} + 0.7937 \right] u[n-1]$$

The result is the same as the result using the  $z$  transform, but the effort and the probability of error are considerably greater.



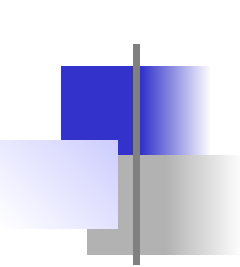
# System Response to a Sinusoid

A system with transfer function

$$H(z) = \frac{z}{z - 0.9}, \quad |z| > 0.9$$

is excited by the sinusoid  $x[n] = \cos(2\pi n / 12)$ . Find the response.

The  $z$  transform of a true sinusoid does not appear in the table of  $z$  transforms. The  $z$  transform of a causal sinusoid of the form  $x[n] = \cos(2\pi n / 12)u[n]$  does appear. We can use the DTFT to find the response to the true sinusoid and the result is  $y[n] = 1.995 \cos(2\pi n / 12 - 1.115)$ .



# System Response to a Sinusoid

Using the  $z$  transform we can find the response of the system to a causal sinusoid  $x[n] = \cos(2\pi n / 12)u[n]$  and the response is

$$y[n] = 0.1217(0.9)^n u[n] + 1.995 \cos(2\pi n / 12 - 1.115)u[n]$$

Notice that the response consists of two parts, a transient response

$0.1217(0.9)^n u[n]$  and a forced response  $1.995 \cos(2\pi n / 12 - 1.115)u[n]$

that, except for the unit sequence factor, is exactly the same as the forced response we found using the DTFT.

# System Response to a Sinusoid

This type of analysis is very common. We can generalize it to say that if a system has a transfer function  $H(z) = \frac{N(z)}{D(z)}$  that the response to a causal cosine excitation  $\cos(\Omega_0 n)u[n]$  is

$$y[n] = \underbrace{Z^{-1}\left[z \frac{N_1(z)}{D(z)}\right]}_{\text{Natural or Transient Response}} + \underbrace{\left|H(p_1)\right| \cos(\Omega_0 n + \angle H(p_1))}_{\text{Forced Response}} u[n]$$

where  $p_1 = e^{j\Omega_0}$ . This consists of a natural or transient response and a forced response. If the system is stable the transient response dies away with time leaving only the forced response which, except for the  $u[n]$  factor is the same as the forced response to a true cosine. So we can use the  $z$  transform to find the response to a true sinusoid.